

CSCI E-106: Section 03 Spring 2025

#You should have solved something similar in the homework but lets see it again! # 1.20. Copier maintenance. The Tri-City Office Equipment Corporation sells an imported copier on a franchise basis and performs preventive maintenance and repair service on this copier. The data below have been collected from 45 recent calls on users to perform routine preventive maintenance service; for each call, X is the number of copiers serviced and Y is the total number of minutes spent by the service person. Assume that first-order regression model (1.1) is appropriate.

#Loading Data

```
# Load our data! Here we are loading a txt file so note the delim
# Assigning the data to a data frame object
# called df20 and then we want to name the columns

df20 = read.delim("CH01PR20.txt", header=FALSE, sep="")
colnames(df20) = c("y", "x")
```

a Obtain the estimated regression function.

Solution

```
#this is our estimated regression function
lmFit20 = lm(y~x, df20)
#lets look at our estimated regression function

summary(lmFit20)

##
## Call:
## lm(formula = y ~ x, data = df20)
##
## Residuals:
```

	Min	1Q	Median	3Q	Max
##	-22.7723	-3.7371	0.3334	6.3334	15.4039

```
##
## Coefficients:
```

	Estimate	Std. Error	t value	Pr(> t)
## (Intercept)	-0.5802	2.8039	-0.207	0.837
## x	15.0352	0.4831	31.123	<2e-16 ***

```
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

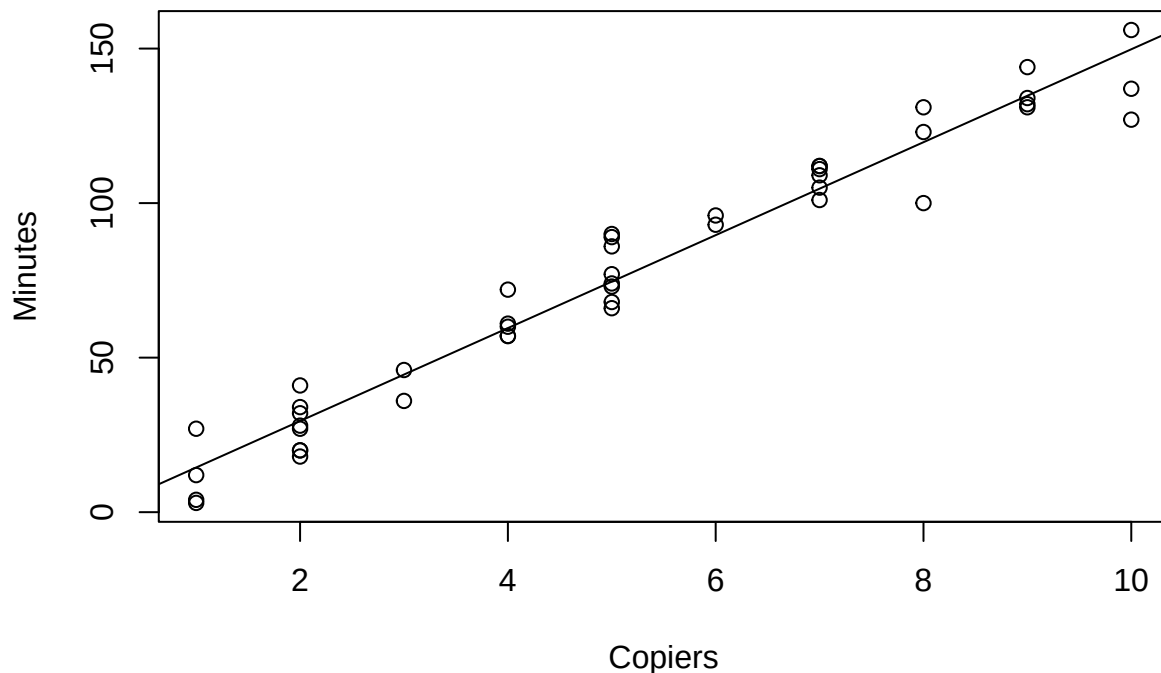
We see that to obtain our estimated regression function we would use the estimate column values so our answer would become: $y = -.5802 + 15.0352x$

b. Plot the estimated regression function and the data. How well does the estimated regression function fit the data?

Solution

```
#here we will plot our estimated regression function
plot(df20$x, df20$y, ylab="Minutes", xlab="Copiers")

#here is our best fit line on our regression. We can look at our outliers & decide if the relationship
abline(lmFit20)
```



By just looking at our data we do have a well fit line here - so it is looking to be linear with no outliers.

c. Write down your standard error of the slope and intercept.

Solution

```
#calculate your standard error
coef(summary(lmFit20))[, "Std. Error"]

## (Intercept)          x
##  2.8039411    0.4830872

#or you can show this with your summary
summary(lmFit20)

##
## Call:
## lm(formula = y ~ x, data = df20)
##
## Residuals:
```

```
##      Min      1Q   Median      3Q      Max
## -22.7723 -3.7371  0.3334   6.3334  15.4039
##
## Coefficients:
##           Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## x             15.0352     0.4831  31.123 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
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```

d. How does our model perform?

Solution

```
library(caret)
```

```
## Loading required package: ggplot2
```

```
## Loading required package: lattice
```

```
##
```

```
## Attaching package: 'lattice'
```

```
## The following object is masked from 'package:faraway':
```

```
##
```

```
##      melanoma
```

```
#we can use predict to check on the future values based on the previous data and with the help of the l
Predicted<-predict(lmFit20,df20)
```

```
#once you predict - you want to see a summary
#the defaultSummary() function in R is used to calculate
#and report standard performance metrics for regression models.
```

```
Model<-data.frame(obs = df20$y, pred=Predicted)
```

```
Model=defaultSummary(Model)
```

```
print(Model)
```

```
##      RMSE  Rsquared      MAE
```

```
## 8.7131791 0.9574955 6.9321265
```

```
#from here we would expect you to look at the results and comment on your findings
```

2.05. Reference to Copier maintenance

The Tri-City Office Equipment Corporation sells an imported copier on a franchise basis and performs preventive maintenance and repair service on this copier. The data below have been collected from 45 recent calls on users to perform routine preventive maintenance service; for each call, X is the number of copiers serviced and Y is the total number of minutes spent by the service person. Assume that first-order regression model (1.1) is appropriate.

```
# Load data
df20 = read.delim("CH01PR20.txt", header=FALSE, sep="")
colnames(df20) = c("y", "x")
```

a. Estimate the change in the mean service time when the number of copiers serviced increases by one. Use a 90 percent confidence interval. Interpret your confidence interval.

```
# Create linear model from data
```

```
lmFit20 = lm(y~x, df20)
```

```
print(summary(lmFit20))
```

```
##
## Call:
## lm(formula = y ~ x, data = df20)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.7723  -3.7371   0.3334   6.3334  15.4039
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -0.5802     2.8039  -0.207   0.837
## x             15.0352     0.4831  31.123 <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 8.914 on 43 degrees of freedom
## Multiple R-squared:  0.9575, Adjusted R-squared:  0.9565
## F-statistic: 968.7 on 1 and 43 DF,  p-value: < 2.2e-16
```

So the estimated change in the mean service time when the number of copiers serviced increases by one is: 15.0352 which is the estimated slope (b_1) and the $s(b_1)$ is .4831

```
#Finding our t-distribution
```

```
#In this case we want to first get our degrees of freedom which we can see from our summary is: 43
```

```
#or we can use our anova function to find the same
```

```
anova(lmFit20)
```

```
## Analysis of Variance Table
```

```
##
```

```
## Response: y
```

```
##           Df Sum Sq Mean Sq F value    Pr(>F)
```

```
## x           1  76960   76960  968.66 < 2.2e-16 ***
```

```
## Residuals 43    3416         79
```

```
## ---
```

```
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
#We can find our t-distribution with our degrees of freedom
```

```
tdist = qt(1-.10/2, 43)
```

```
print(tdist)
```

```
## [1] 1.681071
```

```
#Use a 90 Percent Confidence Interval
```

```
confint(lmFit20, level=.9)
```

```
##              5 %      95 %
```

```
## (Intercept) -5.29378  4.133467
```

```
## x          14.22314 15.847352
```

```
#or you can write it all out without a function:  
c(15.0352-1.6811*0.4831, 15.0352+1.6811*0.4831)
```

```
## [1] 14.22306 15.84734
```

So our b_1 interval would be: $14.223 \leq b_1 \leq 15.847$ 90% of the time. with $t(.95,43)$: 1.681

b. Conduct a t test to determine whether or not there is a linear association between X and Y here; control the a risk at .10. State the alternatives, decision rule, and conclusion. What is the P – value of your test?

$H_0: B_1 = 0, H_a: B_1 \neq 0$

So our test statistic is: $t = b_1 / SD(b_1) = (15.0352 - 0) / .4831$ Or you can see it above under t value of x

```
t = ( 15.0352 - 0 ) / .4831  
print(t)
```

```
## [1] 31.12233
```

The decision rule would be: reject H_0 if $t > 1.681$ or reject H_0 if the p-value $< .1$

The conclusion: Reject the hypothesis

P-Value: $P(t_{43} > 31.123) < 0.000001$ or as seen in our summary: $2.2e-16$

```
#Calculate the p-value  
p = 1-pt(31.122,43)  
print(p)
```

```
## [1] 0
```